

SUPPLEMENT TO CERTIFIED BELLMAN OPERATORS WITH NEURAL PROPOSALS

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1. OBJECTS, ARITHMETIC, AND PRESERVATION

This supplement supplies the analytic construction behind the active-boundary calculation and the detailed numerical records for the regional certificate. The five R29 documents are reproduced, without page alteration, in `HISTORY_R30.pdf`. The original source, exact-rational inventory records, controls, failures, and preceding responses remain in their original directories. Historical statements are to be read with the current response and scope statements, not as an alternative set of current claims.

Three arithmetic objects must remain distinct. Training uses floating point. Compilation interprets the saved binary64 coefficients as exact rational constants and produces an exact piecewise-constant decision rule with explicit tie points. The certificate uses rational polynomial bounds, comparisons, and integrals. An actual floating-point neural forward pass is not certified. The integer-index deployment test covers arbitrary-precision rational comparisons, not unmodeled hardware rounding.

A candidate's saved parameters, compiled intervals, and final cover are linked in the verifier. Every certificate leaf must lie within a compiled constant-action interval and carry the correct raw decision. Every breakpoint is checked against exact network evaluation. Coverage, adjacency, feasible actions, bound coefficients, endpoint actions, and the final tolerance are verified independently of the training loss. Removing an endpoint, altering a bound, making the cover incomplete, supplying a negative tolerance, or querying an invalid grid index is rejected.

2. STOPPED GAUSSIAN MOMENTS AND ANALYTIC ERRORS

Let $a = 6/5$, $b = 14/5$, $u_0 = 61/50$, $\sigma = 1/20$, and $c = 3/4$. Deterministic wealth is

$$x(t) = \left(\frac{5}{4} - \frac{3/4}{.02} \right) e^{.02t} + \frac{3/4}{.02}. \quad (1)$$

It decreases to a value above .51770, so there is no wealth exit in this restricted class. The original preference boundaries and settlement are unchanged. Put $q = y - 2$, so $|q| \leq 4/5$ on $[a, b]$, and $L = \log(4/3)$. The utility term is

$$\frac{c^{1-y}}{1-y} = -\frac{e^{L(1+q)}}{1+q} = \sum_{n=0}^{\infty} c_n q^n, \quad c_n = -\frac{4}{3} \sum_{j=0}^n \frac{L^j}{j!} (-1)^{n-j}. \quad (2)$$

Since $|c_n| \leq e^{2L} = 16/9$, truncation after degree 128 has the uniform bound

$$R_{128} \leq \frac{16}{9} \frac{(4/5)^{129}}{1 - 4/5} < 2.803 \cdot 10^{-12}. \quad (3)$$

All coefficients are enclosed by ball arithmetic. The remainder is an analytic uniform bound, not an interpolation residual.

The half-line killed kernel equals a difference of two Gaussian densities, with means $m_1 = u_0 + \theta t$ and $m_2 = 2a - u_0 + \theta t$, common variance $v = \sigma^2 t$, and image multiplier $e^{-2\theta(u_0 - a)/\sigma^2}$. For a Gaussian density $\varphi_{m,v}$ define

$$I_n(m, v) = \int_a^b (y - 2)^n \varphi_{m,v}(y) dy. \quad (4)$$

The initial moment is a normal-CDF difference. Integration by parts gives

$$I_1 = (m - 2)I_0 + v\{\varphi_{m,v}(a) - \varphi_{m,v}(b)\}, \quad (5)$$

$$I_n = (m - 2)I_{n-1} + (n - 1)vI_{n-2} + v\{(a - 2)^{n-1}\varphi_{m,v}(a) - (b - 2)^{n-1}\varphi_{m,v}(b)\}, \quad n \geq 2. \quad (6)$$

Their image-weighted difference integrates the utility polynomial and the polynomial components of the Dynkin integrand exactly up to arithmetic enclosure. Moments through degree 130 suffice for two parameter derivatives.

In the fixed $k = 2$ slice,

$$h_\theta(t, y) = \frac{c^{1-y}}{1-y} - \theta^2 + 8 - .04\theta(y - 2) - .00005 + .002 - \frac{c}{10x(t)} \quad (7)$$

$$- .04\{-.02(y - 2)^2 + .1 \log x(t)\} + .32(1 - t),$$

$$\partial_\theta h_\theta = -2\theta - .04(y - 2), \quad \partial_\theta^2 h_\theta = -2. \quad (8)$$

On the domain, the deliberately conservative bounds $|h| \leq M = 20$, $|h'| \leq M_1 = 2$, and $|h''| \leq M_2 = 8$ hold. For example, the utility magnitude is below 5.3, wealth is above .5177, and the remaining displayed terms fit strictly within these bounds.

The time interval $[0, t_0]$, with $t_0 = 10^{-6}$, is bounded rather than numerically integrated. Under the full Gaussian law,

$$\mathbb{E}|S_t| \leq \frac{\sqrt{t}}{\sigma}, \quad \mathbb{E}\left|S_t^2 - \frac{t}{\sigma^2}\right| \leq \frac{\sqrt{2}t}{\sigma^2}. \quad (9)$$

Restricting to survival cannot increase these absolute-moment bounds. Thus the three omitted-time errors are at most

$$E_0^{\text{time}} = Mt_0, \quad (10)$$

$$E_1^{\text{time}} = M_1 t_0 + \frac{2M}{3\sigma} t_0^{3/2}, \quad (11)$$

$$E_2^{\text{time}} = M_2 t_0 + \frac{4M_1}{3\sigma} t_0^{3/2} + \frac{M\sqrt{2}}{2\sigma^2} t_0^2. \quad (12)$$

The uniform utility approximation contributes at most R_{128} , $2R_{128}/(3\sigma)$, and $R_{128}\sqrt{2}/(2\sigma^2)$, respectively.

The difference between the true doubly stopped occupation measure and the lower-killed measure restricted to $[a, b]$ occurs only on a path that has already hit the upper boundary. With the probability bound p_b in the article, Cauchy–Schwarz and the likelihood-score second moments

yield opposite-boundary errors

$$E_0^{\text{opp}} = Mp_b, \quad (13)$$

$$E_1^{\text{opp}} = M\sqrt{p_b}/\sigma + M_1p_b, \quad (14)$$

$$E_2^{\text{opp}} = M\sqrt{2p_b}/\sigma^2 + 2M_1\sqrt{p_b}/\sigma + M_2p_b. \quad (15)$$

This treatment does not discard the economically material lower-boundary event. It only encloses the distant upper-boundary correction. Adding these errors to the ball-integrated polynomial expression produces the reported derivative enclosures.

The remaining integration uses $s = \sqrt{t}$, 200-bit arithmetic, and the fixed breakpoints $s \in \{.001, .01, .05, .1, .2, .4, .8, 1\}$. The complex analytic extension passes holomorphicity checks to square root and logarithm; a ball crossing an unverified branch domain returns nonfinite rather than a false bound. The FLINT integration routine supplies a rigorous integral enclosure; its requested decimal tolerance is not itself treated as the achieved error bound. Final balls are tested for finiteness, reality, and radius below .005. Their actual radii are much smaller. The first attempted nested spatial/time ball integral failed; its source and output remain in `results/stopping/attempts`. The successful moment formulation changes the implementation, not the instance or target controls.

3. EXACT COST ACCOUNTING

For a raw-action interval with local deficit bounds $[l_d, u_d]$, the mandatory-change lower bound includes it only when $l_d > \eta$. The reported upper bound includes every changed interval. Their difference therefore bounds the intervention cost in excess of the minimum local-loss-class cost even when the subdivision stops conservatively near a tolerance crossing. The online exact guard attains the same local-class optimum pointwise without constructing a finite cover. The finite cover provides an offline certificate and an alternative deployment representation, not a lower intervention cost than that guard.

Under the continuous reset density, interval occupancy is $\int_l^r 2x \, dx$ and priority-weighted occupancy is $\int_l^r 2x(1+x) \, dx$. Multiplication by H_8 accounts for survival and discounting. Operating gains are integrated from $(a^{\text{new}} - a^0)f(x)$ against that same density. An independently written interval-overlay routine recomputes the policy-change costs and the comparison to the near-exact structural policy. No finite-grid occupancy estimate is substituted for these continuous integrals.

4. DETAILED EVIDENCE

TABLE I: All neural candidates; A = Adam, B = L-BFGS

Gain	Method/width/seed	Raw regret upper	Cells	Calls	Occupancy	Priority	Full s
linear	A/8/1	0.000039	3	3	0	0	0.9121
linear	A/8/2	0.012663	41	77	0.03773	0.0516	0.07108
linear	A/8/3	0.000024	3	3	0	0	0.06399
linear	A/8/4	0.000783	4	4	0	0	0.06383
linear	A/8/5	0.000007	4	4	0	0	0.06447
linear	B/8/1	0.000003	3	3	0	0	0.02075
linear	B/8/2	0.000009	5	5	0	0	0.0221

Gain	Method/width/seed	Raw regret upper	Cells	Calls	Occupancy	Priority	Full s
linear	B/8/3	0.000001	3	3	0	0	0.01558
linear	B/8/4	0.000006	2	2	0	0	0.008753
linear	B/8/5	0.000002	3	3	0	0	0.01174
linear	A/16/1	0.000167	4	4	0	0	0.06609
linear	A/16/2	0.004521	41	77	0.009016	0.01237	0.07411
linear	A/16/3	0.000006	2	2	0	0	0.06553
linear	A/16/4	0.000024	3	3	0	0	0.06516
linear	A/16/5	0.002148	44	80	0.0004114	0.0005666	0.07409
linear	B/16/1	0.000002	6	6	0	0	0.02501
linear	B/16/2	0.000025	6	6	0	0	0.02479
linear	B/16/3	0.000257	6	6	0	0	0.009412
linear	B/16/4	0.000008	5	5	0	0	0.01304
linear	B/16/5	0.000001	7	7	0	0	0.03418
cubic	A/8/1	0.420749	144	286	2.935	4.579	0.127
cubic	A/8/2	0.071483	115	223	0.2479	0.3731	0.1147
cubic	A/8/3	0.384982	110	217	2.387	3.79	0.113
cubic	A/8/4	0.384866	110	217	2.386	3.789	0.1169
cubic	A/8/5	0.420749	144	286	2.935	4.579	0.1844
cubic	B/8/1	0.420749	144	286	2.935	4.579	0.06817
cubic	B/8/2	0.030503	80	152	0.09845	0.1649	0.1278
cubic	B/8/3	0.385395	110	217	2.39	3.795	0.06425
cubic	B/8/4	0.385395	110	217	2.39	3.795	0.06486
cubic	B/8/5	0.385392	110	217	2.39	3.795	0.06136
cubic	A/16/1	0.384537	111	218	2.383	3.785	0.1165
cubic	A/16/2	0.403565	110	217	2.562	4.034	0.1108
cubic	A/16/3	0.384208	112	219	2.38	3.781	0.1208
cubic	A/16/4	0.045918	115	223	0.1029	0.1659	0.1229
cubic	A/16/5	0.018225	47	83	0.04472	0.08034	0.09018
cubic	B/16/1	0.032401	116	224	0.1247	0.1975	0.1729
cubic	B/16/2	0.032752	115	223	0.1425	0.2324	0.09924
cubic	B/16/3	0.383961	112	219	2.378	3.779	0.09589
cubic	B/16/4	0.032828	115	223	0.1442	0.2349	0.1377
cubic	B/16/5	0.029961	117	225	0.09242	0.1585	0.1801
quadratic	A/8/1	0.059549	73	144	0.8961	1.209	0.08238
quadratic	A/8/2	0.001982	6	6	0	0	0.06422
quadratic	A/8/3	0.059549	74	145	0.8961	1.209	0.08016
quadratic	A/8/4	0.058462	77	149	0.5674	0.7913	0.08809
quadratic	A/8/5	0.059549	74	146	0.8961	1.209	0.08306
quadratic	B/8/1	0.059549	73	144	0.8961	1.209	0.02752
quadratic	B/8/2	0.014738	77	149	0.1417	0.1945	0.05122
quadratic	B/8/3	0.010987	78	150	0.07094	0.104	0.0445
quadratic	B/8/4	0.015021	77	149	0.1403	0.1922	0.0612
quadratic	B/8/5	0.203274	74	146	0.9891	1.307	0.02883
quadratic	A/16/1	0.005034	78	150	0.02653	0.03837	0.08954
quadratic	A/16/2	0.059175	78	150	0.8797	1.186	0.08849
quadratic	A/16/3	0.046384	77	149	0.4816	0.6568	0.0917
quadratic	A/16/4	0.004746	78	150	0.01232	0.01552	0.09207
quadratic	A/16/5	0.024038	77	149	0.1023	0.1259	0.08828
quadratic	B/16/1	0.006289	79	151	0.03104	0.04194	0.134
quadratic	B/16/2	0.005631	79	151	0.02927	0.03989	0.1475
quadratic	B/16/3	0.005626	79	151	0.02757	0.0374	0.156
quadratic	B/16/4	0.021356	42	78	0.1371	0.2043	0.05929
quadratic	B/16/5	0.004861	80	152	0.02913	0.04057	0.2042

TABLE II
POLYNOMIAL CONTROLS WITH THE SAME OBSERVATIONS AND KNOWN DEGREE

Gain	Raw regret upper	Fit s	Compile s	Certify s	Cells
linear	0E-12	0.002705	0.001906	0.000161	2
cubic	1E-12	0.0001478	0.01762	0.0008435	4
quadratic	1E-12	0.0002166	0.006903	0.0002222	3

These are exact compiled dyadic policies from fitted polynomial coefficients. An irrational root is not asserted to equal its dyadic approximation.

5. SCOPE OF THE RETAINED CONTINUOUS AND HISTORICAL RESULTS

The scalar comparative statics, the finite inventory completion, the regional operating-mode certificate, and the original continuous all-domain objective have different policy classes and restart sets. The new active-boundary calculation establishes payoff and derivative enclosures for five constant controls at one restart. It neither supplies a global scalar optimizer nor validates all 47 financed coordinates. The original bound 7.181834580823298 is unchanged. The original supersolution/residual theorem and every earlier unsuccessful run remain available in the reproduced R29 documents.

The central-state bracketing comparison uses the inherited validated oracle without changing its error budget or strict-concavity audit. Its target is a projected-gradient upper bound .005; the inherited polling schedule reports its own achieved bound, which need not be identical. Setup and concavity-audit costs are charged before either search. The old direct-search theorem is retained as a sufficient certificate theorem, not as evidence that polling dominates root-based scalar methods. The historical transport identity and restoration tests are preserved in full but no longer serve as independent numerical-superiority claims.

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